

Math 421

Monday, September 29

Announcements

- Homework 4 due Friday
- Grace's Drop-In Hours
 - Wednesday: 12-1 VV 311
 - Thursday: 9-11 VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4

Convergence

Def: A sequence (s_n) of real numbers is said to **converge** to the real number s provided that:

For each $\epsilon > 0$ there exists a number N such that
whenever $n > N$ implies $|s_n - s| < \epsilon$.

$\hookrightarrow$ solve for $n > \boxed{}$ "N"

We can think of the definition of convergence as a game:
someone gives you (a very small) $\epsilon > 0$ and you need to
find some N that makes the statement “whenever $n > N$,
it follows that $|s_n - s| < \epsilon$ ” true.

Group Activity – Divide & Conquer

Form groups of 3-6 and further split into 3 subgroups (A, B, C). Each subgroups should find the appropriate N for their specific ϵ value in each of the following examples. Then compare as a full group and answer the discussion question.

1. $s_n = \frac{1}{n}, s = 0, \epsilon = 1, \frac{1}{10}, \frac{1}{100}$

$$N = \frac{1}{\epsilon}$$

$$n > N$$

$$|s_n - s| < \epsilon$$

2. $s_n = \frac{1}{e^n}, s = 0, \epsilon = 1, \frac{1}{10}, \frac{1}{100}$

$$N = \ln\left(\frac{1}{\epsilon}\right)$$

Q: Can you generalize what you found to obtain an expression for N given an arbitrary $\epsilon > 0$, for each example?

Recipe to show that $\lim s_n = s$

Step 0: Do scratch work to determine what N should be.

Step 1: Fix an arbitrary $\epsilon > 0$ and state the explicit expression for N , that you determined in your scratch work.

Step 2: Show that for all $n \geq N$, $|s_n - s| < \epsilon$. This often follows the algebra from your scratchwork.

Step 3: State that since $\epsilon > 0$ was arbitrary, you have proven, from the definition, that $\lim s_n = s$.

Example: Show that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$

Discussion.

$$\left| \frac{1}{n} - 0 \right| < \varepsilon \iff \left| \frac{1}{n} \right| < \varepsilon \iff \frac{1}{n} < \varepsilon \iff n > \frac{1}{\varepsilon}$$

Proof. Let $\varepsilon > 0$. Set $N = \frac{1}{\varepsilon}$. Assume $n > N$, then

$$\left| \frac{1}{n} - 0 \right| = \left| \frac{1}{n} \right| = \frac{1}{n} < \frac{1}{N} = \frac{1}{\frac{1}{\varepsilon}} = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

Example: Show that $\lim \frac{n^2+1}{n^2} = 1$

Discussion.

$$\left| \frac{n^2+1}{n^2} - 1 \right| = \left| \frac{n^2+1-n^2}{n^2} \right| = \left| \frac{1}{n^2} \right| < \varepsilon \iff$$
$$\frac{1}{n^2} < \varepsilon \iff n^2 > \frac{1}{\varepsilon} \iff n > \sqrt{\frac{1}{\varepsilon}}$$

Activity – Write the Proof.

Activity: Show that $\lim \frac{1}{\sqrt{n}} = 0$

Discussion.

$$N = \frac{1}{\epsilon^2}$$

Proof.

$$\left| \frac{1}{\sqrt{n}} - 0 \right| = \dots \dots \dots < \epsilon$$

Example

for all $\varepsilon > 0$, there exists N s.t.

$n > N$

$$|s_n - s| < \varepsilon \cdot \sqrt{s}$$

solve for this

Theorem. If $\lim s_n = s$, $s_n \geq 0$ for all n , and $s > 0$, then $\lim \sqrt{s_n} = \sqrt{s}$.

Discussion. $|\sqrt{s_n} - \sqrt{s}| < \varepsilon$.

appropriate representation of 1

$$|\sqrt{s_n} - \sqrt{s}| = \left| \frac{(\sqrt{s_n} - \sqrt{s})(\sqrt{s_n} + \sqrt{s})}{(\sqrt{s_n} + \sqrt{s})} \right| = \frac{|s_n - s|}{\sqrt{s_n} + \sqrt{s}} < \frac{|s_n - s|}{\sqrt{s}}$$

$$\Leftrightarrow \frac{|s_n - s|}{\sqrt{s}} < \varepsilon$$

$$\Leftrightarrow |s_n - s| < \sqrt{s} \varepsilon$$